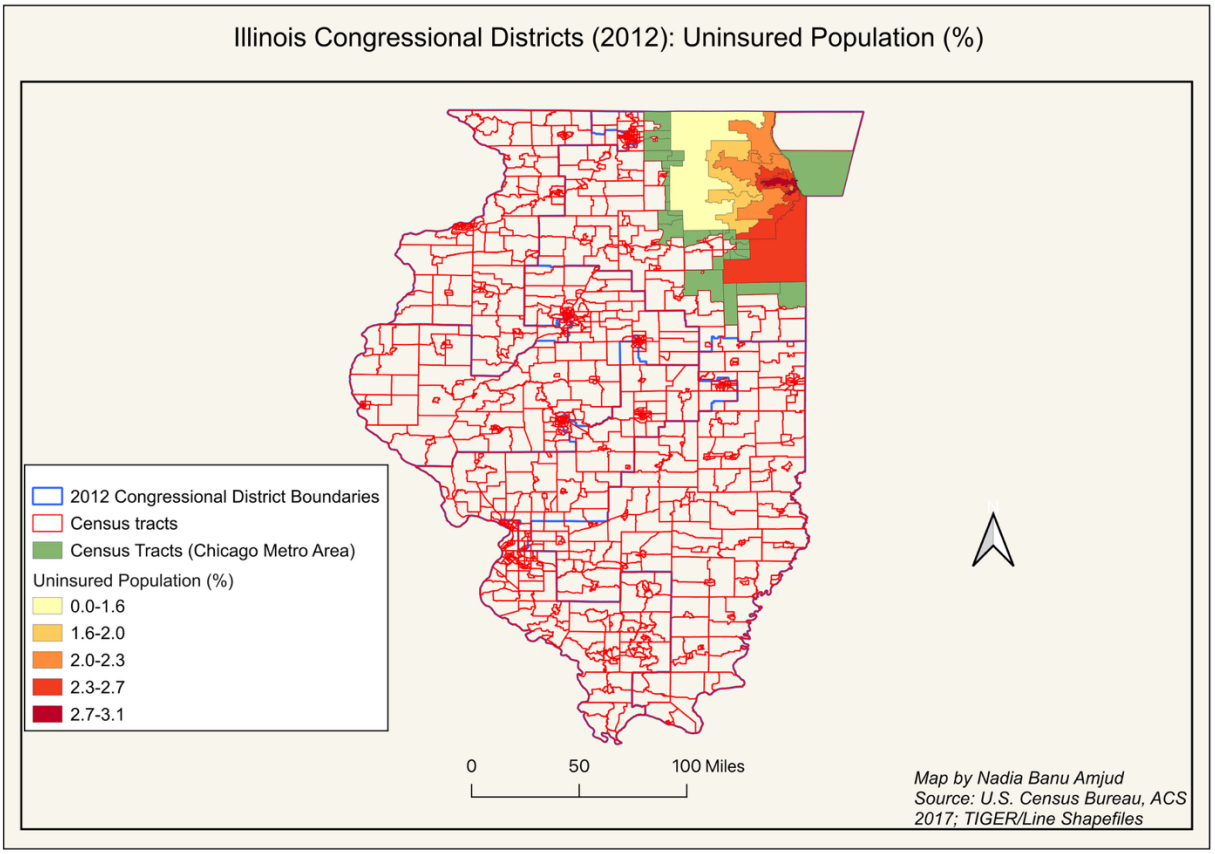
Project 1: Voting & Demography in Illinois

Overview

This project examines the relationship between congressional voting patterns and demographic characteristics in Illinois using post-2012 redistricting data. Census tract level socioeconomic indicators were integrated with congressional district boundaries through multiple spatial join methods. Final outputs included comparative join layers and a choropleth map visualizing health-insurance disparities across districts.

Spatial Skills Used

This project applied key GIS techniques, including attribute joins, spatial selections, and two types of spatial joins (polygon intersection and centroid-based) to link Census tract demographics with congressional districts. I performed data normalization and created calculated fields (e.g., uninsured population percentage) using the Field Calculator. I produced a graduated choropleth map using effective classification schemes, color ramps, and layout design elements. The project demonstrates proficiency in integrating tabular and spatial datasets, summarizing multi-scale geographies, and creating clear, policy-relevant map visualizations.



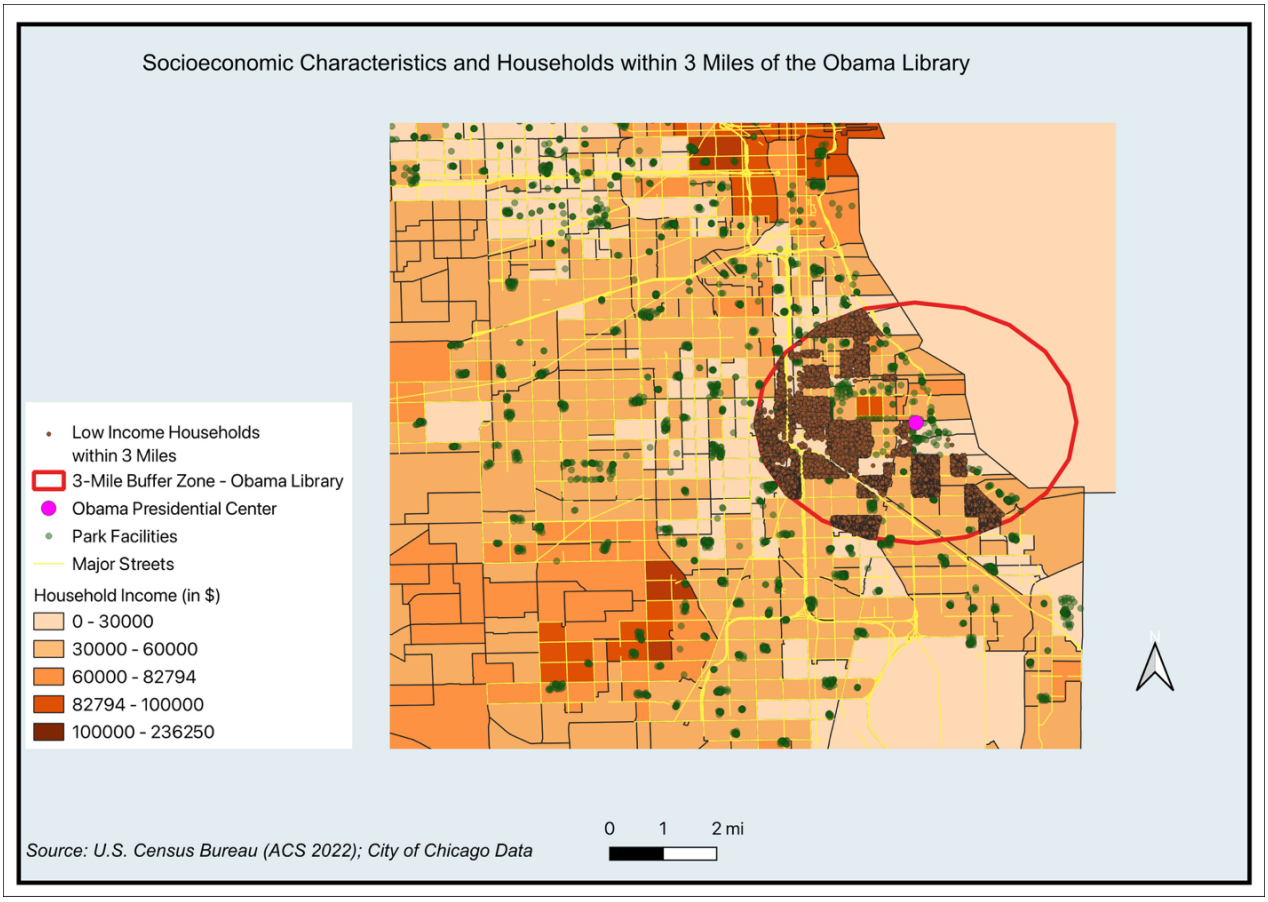
Project 2: Socioeconomic Characteristics and Low-Income Households Near the Obama Presidential Center

Overview

This project maps the spatial distribution of low-income households within 3- and 5-mile areas surrounding the Obama Presidential Center on Chicago's South Side. The analysis integrates ACS demographic data, geocoded household address points, and contextual layers such as parks and streets. Buffering, querying, clipping, and overlay analysis were used to compare the concentration of low-income households within and beyond the buffer zones.

Spatial Skills Used

I applied a wide range of GIS techniques, including importing and geoprocessing CSV address data, assigning and managing coordinate reference systems, and digitizing new point features for key locations. Spatial analysis operations included buffering, attribute queries, and clipping household points and tracts. Thematic mapping was accomplished using graduated symbology of ACS income data, and final layouts incorporated professional cartographic elements such as legends, scale bars, visual hierarchy, and source citations. This project highlights strong competency in geoprocessing, spatial querying, and map design in QGIS.



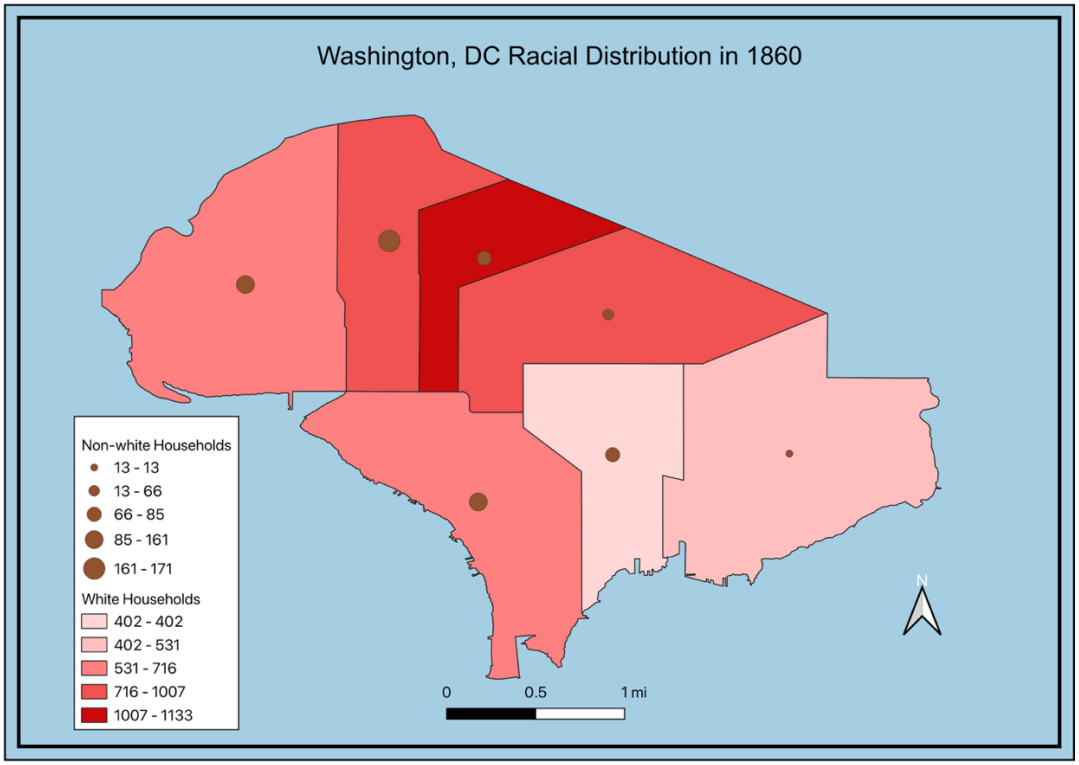
Project 3: Racial Segregation in Washington, DC (1860) and Chicago (2023)

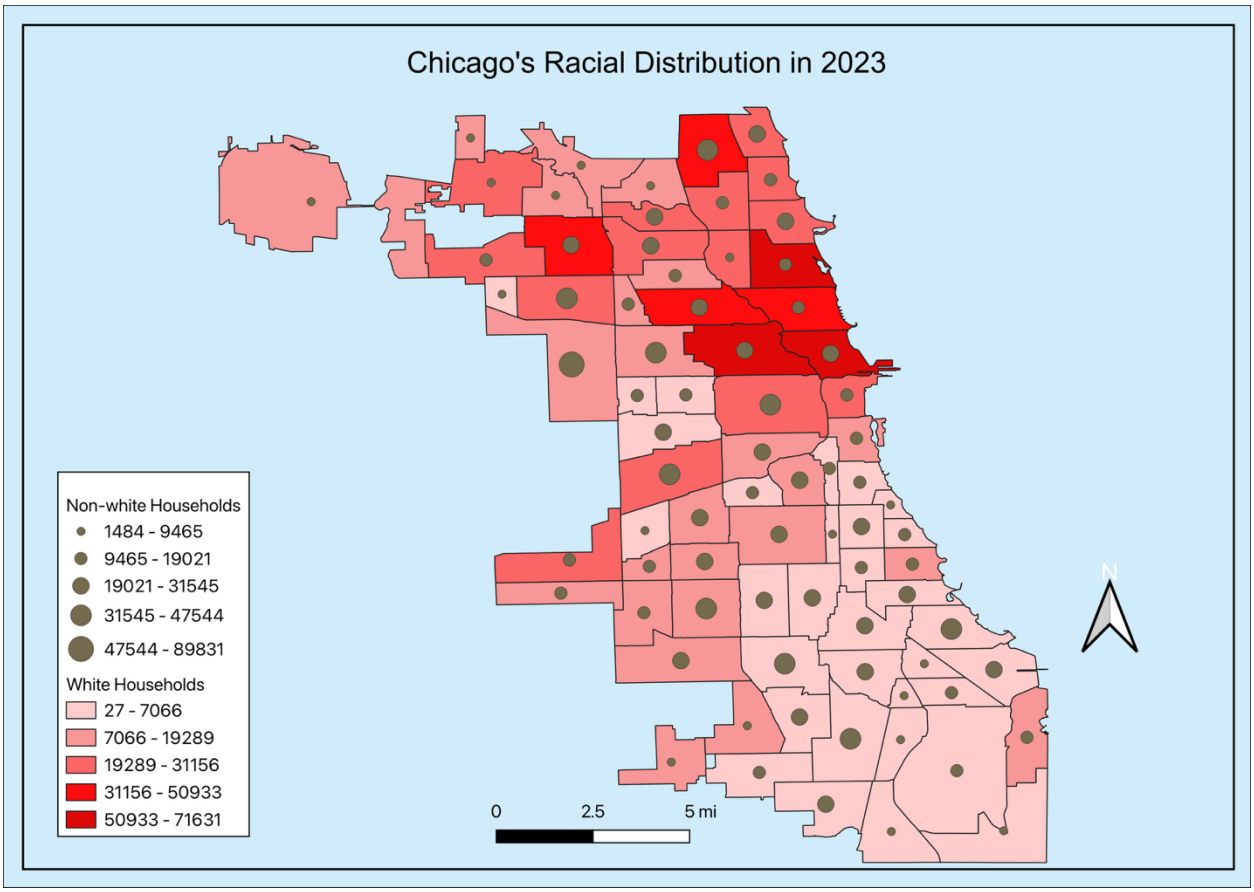
Overview

This project analyzes racial residential patterns across time and place using historical microdata for Washington, DC (1860) and contemporary ACS data for Chicago (2023). I created two maps illustrating racial distribution using QGIS, aggregated demographic attributes through spatial joins, and computed segregation measures including the Dissimilarity Index and the Isolation Index to compare levels of racial unevenness and clustering. The project showcases my ability to integrate spatial analysis, quantitative metrics, and visual interpretation to study urban inequality.

Spatial Skills Used

I performed a variety of GIS and demographic processing tasks, including importing shapefiles, joining demographic tables to spatial boundaries, and conducting point-in-polygon spatial joins to summarize household characteristics. I created new calculated variables (e.g., Nonwhite population), exported attribute tables for statistical analysis, and computed segregation indices in Excel. Map outputs used choropleth classification and graduated symbol mapping with clear legends and color schemes. This work demonstrates proficiency in spatial data management, projection handling, QGIS workflows, and cartographic communication, as well as the integration of spatial visualization with quantitative urban analysis.





Project 4: Spatiotemporal Analysis of Homicide Patterns in Mexico (2007–2010)

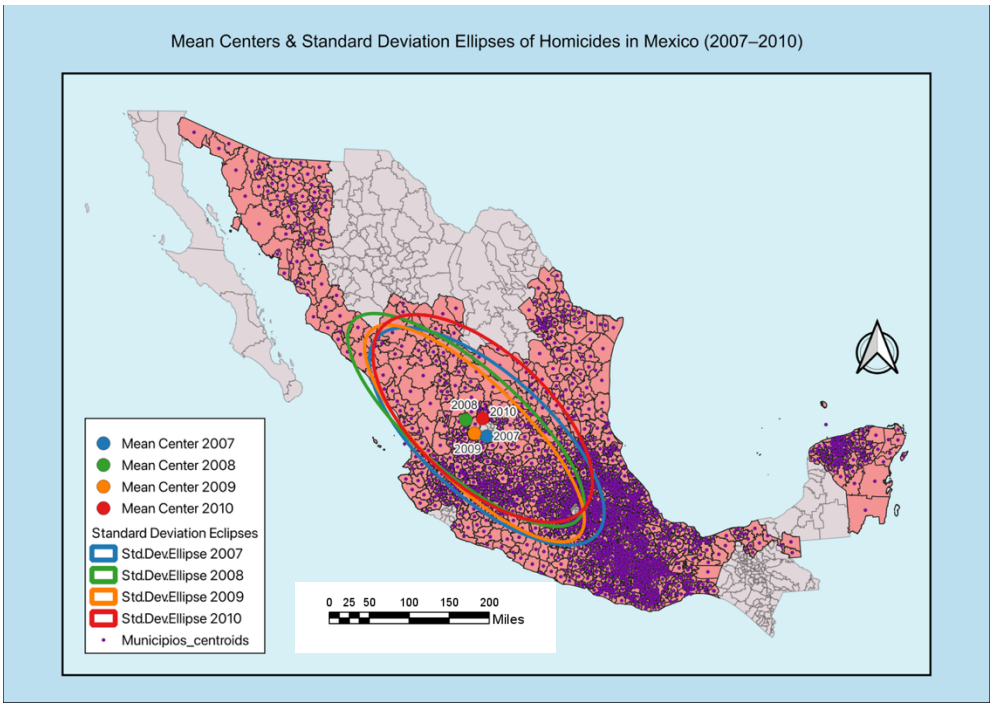
Overview

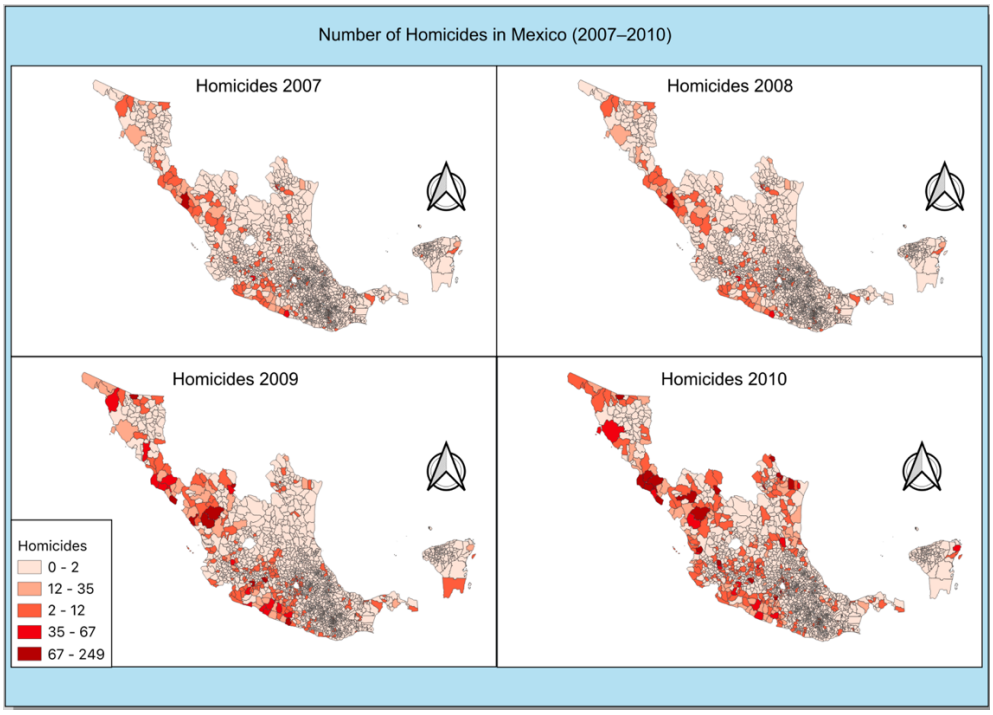
This project analyzes the spatial and temporal dynamics of homicide across Mexico from 2007 to 2010 by integrating municipal-level homicide and education data with national boundary shapefiles. I produced multi-year choropleth visualizations, calculated central tendency and dispersion measures, and applied Local Moran's I (LISA) to identify statistically significant hotspots. Additional regression analysis explored the relationship between homicide rates and educational attainment using GeoDa.

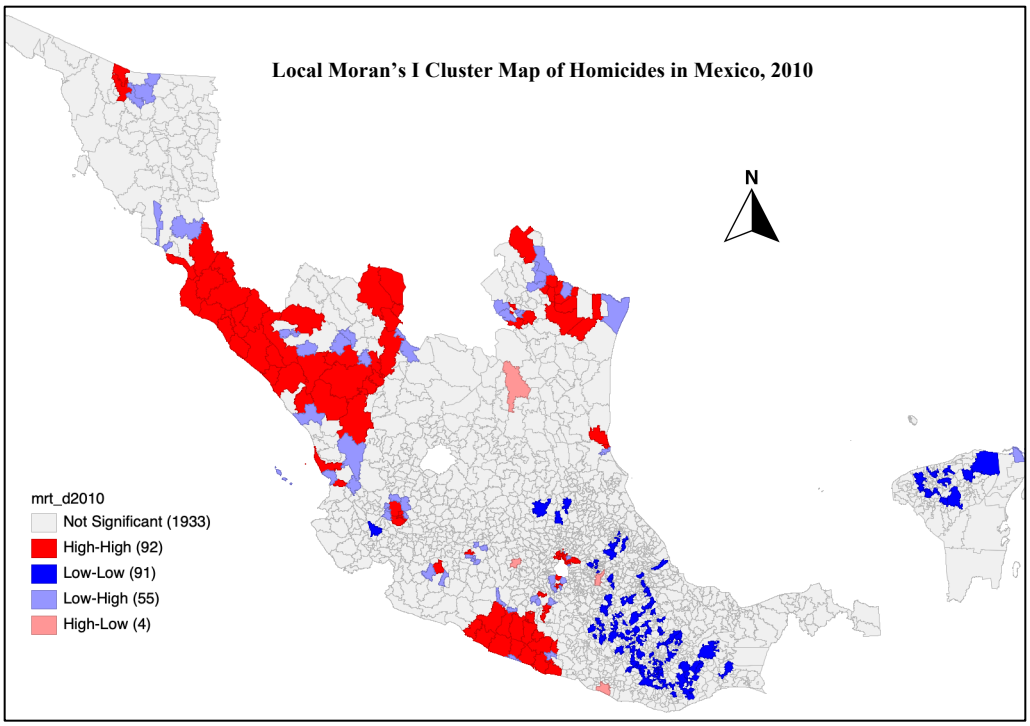
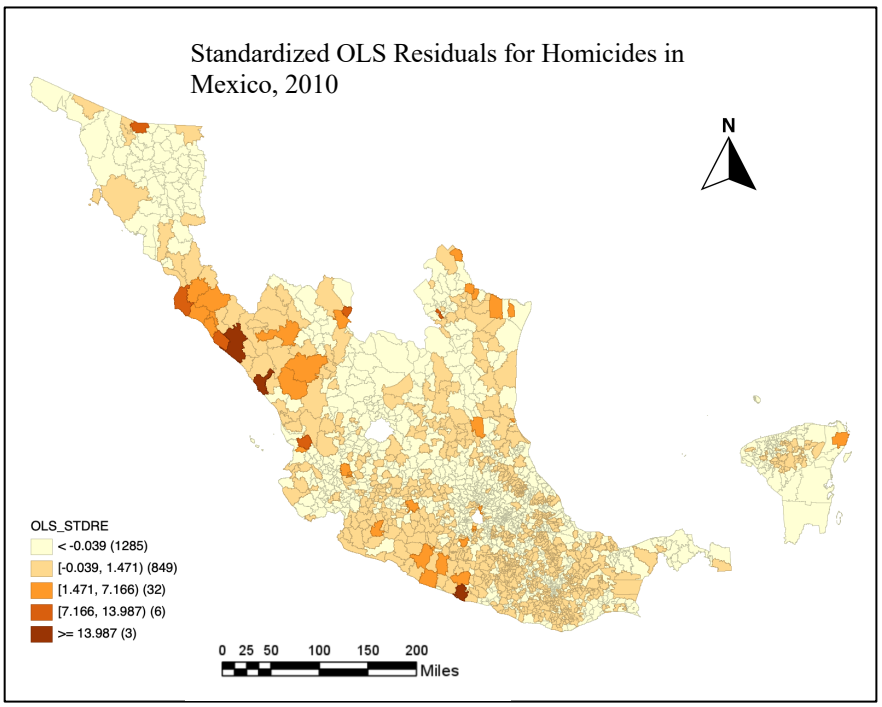
Spatial Skills Used

This project demonstrates advanced spatial analysis skills using QGIS, Python, and GeoDa. I performed spatial joins and created multi-year thematic maps to capture temporal change. Using custom Python scripting, I generated weighted mean centers and standard deviation ellipses to reveal shifting spatial patterns of violence. LISA cluster analysis allowed detection of localized hotspots and spatial autocorrelation. Through consistent classification schemes, strong cartographic design, and clear layout composition, I produced interpretable maps that illustrate both concentration and diffusion of violence over time.

Pattern of violence change in Mexico using Mean Center and Standard Deviation Ellipses







OLS (used GeoDa)